

IMO(26th) – 1985

First Day

1. A circle whose centre is on the side ED of the cyclic quadrilateral BCDE touches the other three sides. Prove that $EB + CD = ED$.
2. Each of the number in the set $N = \{1, 2, 3, \dots, n-1\}$, where $n \geq 3$, is colored with one of the two colors, say red or black, so that:
 - (a) i and $n-i$ always receive the same color;
 - (b) For some $j \in N$ relatively prime to n , i and $|j-i|$ receive the same color for all $i \in N, i \neq j$.
3. The weight $w(p)$ of a polynomial p , $p(x) = \sum_{i=0}^n a_i x^i$, with integer coefficients a_i is defined as the number of its odd coefficients. For $i = 0, 1, 2, \dots$, let $q_i(x) = (1+x)^i$. Prove that for any finite sequence $0 \leq i_1 < i_2 < \dots < i_n$ the inequality $w(q_{i_1} + \dots + q_{i_n}) \geq w(q_{i_1})$ holds.

Second Day

4. Given a set M of 1985 positive integers, none of which has a prime divisor larger than 26, prove that M has four distinct elements whose geometric mean is an integer.
5. A circle with centre O passes through points A and C and intersects the sides AB and BC of the triangle ABC at point K and N respectively. The circumscribed circles of the triangles ABC and KBN intersect at two points B and M. Prove that $\angle OMB = 90^\circ$.
6. The sequence $f_1, f_2, \dots, f_n$ of functions is defined for $x > 0$ recursively by $f_1(x) = x, f_{n+1}(x) = f_n(x) \left(f_n(x) + \frac{1}{n} \right)$. Prove that there exists one and only one positive number a such that $0 < f_n(a) < f_{n+1}(a) < 1$ for all integers $n \geq 1$

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